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2 Quantum Fields

Exercise 2.1

The partition function is

$$\begin{aligned} Z &= \sum_n e^{-\beta\hbar\omega(n+\frac{1}{2})} \\ &= e^{-\beta\hbar\omega/2} \frac{1}{1 - e^{-\beta\hbar\omega}}. \end{aligned} \quad (2.1.1)$$

Let $x = \beta\hbar\omega$,

$$Z = e^{-x/2} (1 - e^{-x})^{-1}. \quad (2.1.2)$$

The internal energy is

$$\begin{aligned} E &= N_A \left(-\frac{\partial \ln Z}{\partial \beta} \right) \\ &= -N_A \hbar\omega \frac{1}{Z} \frac{\partial Z}{\partial x} \\ &= \frac{1}{2} N_A \hbar\omega \left(1 + \frac{e^{-x/2}}{\sinh \frac{x}{2}} \right). \end{aligned} \quad (2.1.3)$$

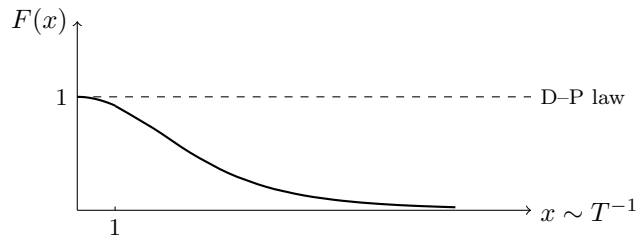
The heat capacity is

$$\begin{aligned} C_V &= \frac{\partial E}{\partial T} = \frac{dx}{dT} \frac{\partial E}{\partial x} \\ &= \frac{1}{4} R x^2 \frac{1}{\sinh^2(\frac{x}{2})} \\ &= R F(x) \end{aligned} \quad (2.1.4)$$

where

$$F(x) = \left(\frac{x/2}{\sinh(x/2)} \right)^2 \rightarrow 1 \quad (2.1.5)$$

in the $x \rightarrow 0$ limit ($k_B T \gg \hbar\omega$), where it recovers the Dulong–Petit law.



Exercise 2.2

$$\begin{aligned} \sum_j \langle q_m | j \rangle \langle j | q_n \rangle &= \frac{1}{N_S} \sum_{j=0}^{N_S-1} e^{i(q_n - q_m)ja} \\ &= \frac{1}{N_S} \sum_{j=0}^{N_S-1} e^{i(n-m)\frac{2\pi}{N_S}j}. \end{aligned} \quad (2.2.1)$$

If $n \neq m$, then¹

$$\begin{aligned}
 \text{LHS} &= \frac{1}{N_S} \frac{e^{i2\pi(n-m)} - 1}{e^{i2\pi(n-m)/N_S} - 1} \\
 &= \frac{1}{N_S} \frac{\sin(\pi(n-m))}{\sin\left(\frac{\pi}{N_S}(n-m)\right)} \frac{e^{-i\pi(n-m)/N_S}}{e^{-i\pi(n-m)}} \\
 &= \frac{1}{N_S} \frac{\sin(\pi(n-m))}{\sin\left(\frac{\pi}{N_S}(n-m)\right)} (-1)^{\frac{N_S-1}{N_S}(n-m)} \\
 &= 0.
 \end{aligned} \tag{2.2.2}$$

If $n = m$, then

$$\begin{aligned}
 \text{LHS} &= \frac{1}{N_S} \sum_{j=0}^{N_S-1} 1 \\
 &= 1.
 \end{aligned} \tag{2.2.3}$$

Therefore,

$$\langle q_m | q_n \rangle = \delta_{mn}. \tag{2.2.4}$$

Exercise 2.3

Have

$$\hat{x}(0) = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger) \tag{2.3.1}$$

$$\hat{x}(t) = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a}e^{i\omega t} + \hat{a}^\dagger e^{-i\omega t}). \tag{2.3.2}$$

The symmetrised correlation function can be expanded as

$$\begin{aligned}
 S_\beta(t) &= \frac{1}{2} \langle \hat{x}(t)\hat{x}(0) + \hat{x}(0)\hat{x}(t) \rangle \\
 &= \frac{\hbar}{4m\omega} [2e^{i\omega t} \langle \hat{a}\hat{a} \rangle + 2e^{-i\omega t} \langle \hat{a}^\dagger \hat{a}^\dagger \rangle + 2 \cos \omega t (\langle \hat{a}\hat{a}^\dagger \rangle + \langle \hat{a}^\dagger \hat{a} \rangle)].
 \end{aligned} \tag{2.3.3}$$

At (inverse) temperature β ,

$$\begin{cases} \langle \hat{a}^\dagger \hat{a} \rangle = \langle \hat{n} \rangle \\ \langle \hat{a}\hat{a}^\dagger \rangle = \langle \hat{a}^\dagger \hat{a} + 1 \rangle = \langle \hat{a} \rangle + 1 \\ \langle \hat{a}\hat{a} \rangle = \langle \hat{a}^\dagger \hat{a}^\dagger \rangle = 0, \end{cases} \tag{2.3.4}$$

where

$$\langle \hat{n} \rangle = n(\beta) = \frac{1}{e^{\beta\hbar\omega} - 1}. \tag{2.3.5}$$

Substituting in (2.3.3) gives

$$S_\beta(t) = \frac{\hbar}{2m\omega} \cos(\omega t) \left[\underbrace{2n(\beta)}_{\text{finite-}T \text{ excitation}} + \underbrace{1}_{\text{zero-point fluctuation}} \right]. \tag{2.3.6}$$

¹I believe the stated result is missing the red N_S^{-1} .

Exercise 2.4

(a)

$$[\hat{b}, \hat{b}] = u^2[\hat{a}, \hat{a}] + uv[\hat{a}, \hat{a}^\dagger] + uv[\hat{a}^\dagger, \hat{a}] + v^2[\hat{a}^\dagger, \hat{a}^\dagger] = 0 \quad (2.4.1)$$

$$[\hat{b}^\dagger, \hat{b}^\dagger] = u^2[\hat{a}^\dagger, \hat{a}^\dagger] + uv[\hat{a}^\dagger, \hat{a}] + uv[\hat{a}, \hat{a}^\dagger] + v^2[\hat{a}, \hat{a}] = 0 \quad (2.4.2)$$

$$[\hat{b}, \hat{b}^\dagger] = u^2[\hat{a}, \hat{a}^\dagger] + uv[\hat{a}, \hat{a}] + uv[\hat{a}^\dagger, \hat{a}^\dagger] + v^2[\hat{a}^\dagger, \hat{a}] = u^2 - v^2. \quad (2.4.3)$$

The first two canonical commutation relations of $\hat{b}$ are guaranteed by the canonical commutation relations of $\hat{a}$, while the last one satisfies $[\hat{b}, \hat{b}^\dagger] = 1$ if and only if $u^2 - v^2 = 1$.

This means that it is convenient to parameterise u, v as $u = \cosh \theta$, $v = \sinh \theta$.

(b) Given $u^2 - v^2 = 1$, the inverse transform is

$$\hat{a} = u\hat{b} - v\hat{b}^\dagger\hat{a}^\dagger = u\hat{b}^\dagger - v\hat{b}. \quad (2.4.4)$$

Then we can expand

$$\begin{aligned} \hat{a}^\dagger\hat{a} + \frac{1}{2} &= u^2\hat{b}^\dagger\hat{b} - uv\hat{b}^\dagger\hat{b}^\dagger - uv\hat{b}\hat{b} + v^2\hat{b}^\dagger\hat{b}^\dagger + \frac{1}{2} \\ &= (u^2 + v^2) \left(\hat{b}^\dagger\hat{b} + \frac{1}{2} \right) - uv(\hat{b}^\dagger\hat{b}^\dagger + \hat{b}\hat{b}) \end{aligned} \quad (2.4.5)$$

and similarly

$$\hat{a}\hat{a} + \hat{a}^\dagger\hat{a}^\dagger = (u^2 + v^2)(\hat{b}\hat{b} + \hat{b}^\dagger\hat{b}^\dagger) - 4uv \left(\hat{b}^\dagger\hat{b} + \frac{1}{2} \right) \quad (2.4.6)$$

using $u^2 - v^2 = 1$. Therefore,

$$\hat{H} = [\omega(u^2 + v^2) - 2\Delta uv] \left(\hat{b}^\dagger\hat{b} + \frac{1}{2} \right) + \left[\frac{\Delta}{2}(u^2 + v^2) - \omega uv \right] (\hat{b}^\dagger\hat{b}^\dagger + \hat{b}\hat{b}). \quad (2.4.7)$$

For $\hat{H}$ to be diagonalisable, there must be some θ such that for $u = \cosh \theta$, $v = \sinh \theta$, the latter term vanish,

$$\Delta(u^2 + v^2) = 2\omega uv. \quad (2.4.8)$$

In θ , this is

$$\Delta \cosh 2\theta = \omega \sinh 2\theta, \quad (2.4.9)$$

so

$$\tanh 2\theta = \frac{\Delta}{\omega}. \quad (2.4.10)$$

Therefore, we must have

$$|\Delta| < \omega \quad (2.4.11)$$

for $\hat{H}$ to be diagonalisable.

The new frequency is

$$\begin{aligned} \tilde{\omega} &= \omega(u^2 + v^2) - 2\Delta uv \\ &= \omega \cosh 2\theta - \Delta \sinh 2\theta. \end{aligned} \quad (2.4.12)$$

Given $\tanh 2\theta = \frac{\Delta}{\omega}$, we have

$$\begin{cases} \sinh 2\theta = \frac{\Delta}{\sqrt{\omega^2 - \Delta^2}} \\ \cosh 2\theta = \frac{\omega}{\sqrt{\omega^2 - \Delta^2}}, \end{cases} \quad (2.4.13)$$

so

$$\tilde{\omega} = \sqrt{\omega^2 - \Delta^2}. \quad (2.4.14)$$

Since

$$\begin{cases} u^2 + v^2 = \cosh 2\theta = \frac{\omega}{\tilde{\omega}} \\ u^2 - v^2 = 1, \end{cases} \quad (2.4.15)$$

we have

$$\begin{cases} u^2 = \frac{1}{2} \left(\frac{\omega}{\tilde{\omega}} + 1 \right) \\ v^2 = \frac{1}{2} \left(\frac{\omega}{\tilde{\omega}} - 1 \right). \end{cases} \quad (2.4.16)$$

We need to take care of the signs when taking the square roots.

$$\begin{cases} u = \sqrt{\frac{1}{2} \left(\frac{\omega}{\sqrt{\omega^2 - \Delta^2}} + 1 \right)} \\ v = \text{sgn}(\Delta) \sqrt{\frac{1}{2} \left(\frac{\omega}{\sqrt{\omega^2 - \Delta^2}} - 1 \right)}. \end{cases} \quad (2.4.17)$$

(c) We have

$$e^{-\alpha \hat{a}^\dagger \hat{a}^\dagger} = \sum_{k=0}^{\infty} \frac{1}{k!} (-\alpha \hat{a}^\dagger \hat{a}^\dagger)^k. \quad (2.4.18)$$

The commutation relations are

$$[\hat{a}^\dagger, e^{-\alpha \hat{a}^\dagger \hat{a}^\dagger}] = 0 \quad (2.4.19)$$

and

$$\begin{aligned} [\hat{a}, e^{-\alpha \hat{a}^\dagger \hat{a}^\dagger}] &= \sum_{k=0}^{\infty} \frac{1}{k!} (-\alpha)^k [\hat{a}, (\hat{a}^\dagger)^{2k}] \\ &= \sum_{k=1}^{\infty} \frac{1}{k!} (-\alpha)^k \cdot 2k (\hat{a}^\dagger)^{2k-1}, \end{aligned} \quad (2.4.20)$$

where in evaluating $[\hat{a}, (\hat{a}^\dagger)^n]$, notice that when you move $\hat{a}$ to the right of a string of $\hat{a}^\dagger$'s, each time you move through an $\hat{a}^\dagger$, you get a $[\hat{a}, \hat{a}^\dagger] = 1$ multiplied with the other $(n-1)$ $\hat{a}^\dagger$'s.

Therefore,

$$\begin{aligned} \hat{b} |\tilde{0}\rangle &= (u\hat{a} + v\hat{a}^\dagger) e^{-\alpha \hat{a}^\dagger \hat{a}^\dagger} |0\rangle \\ &= \left[u \left(e^{-\alpha \hat{a}^\dagger \hat{a}^\dagger} \hat{a} + [\hat{a}, e^{-\alpha \hat{a}^\dagger \hat{a}^\dagger}] \right) + v e^{-\alpha \hat{a}^\dagger \hat{a}^\dagger} \hat{a}^\dagger \right] |0\rangle \\ &= \left\{ u \sum_{k=1}^{\infty} \frac{1}{k!} (-\alpha)^k 2k (\hat{a}^\dagger)^{2k-1} + v \sum_{k=0}^{\infty} \frac{1}{k!} (-\alpha)^k (\hat{a}^\dagger)^{2k+1} \right\} |0\rangle \\ &= \sum_{k=0}^{\infty} \frac{1}{k!} (-\alpha)^k \left[-\frac{\alpha}{k+1} \cdot 2(k+1)u + v \right] (\hat{a}^\dagger)^{2k+1} |0\rangle. \end{aligned} \quad (2.4.21)$$

For this to be 0, each term in the sum must be zero, and hence

$$\alpha = \frac{v}{2u}. \quad (2.4.22)$$

Alternatively: suppose

$$|\tilde{0}\rangle = \sum_{n=0}^{\infty} c_n |n\rangle. \quad (2.4.23)$$

Then

$$\begin{aligned} \hat{b} |\tilde{0}\rangle &= (u\hat{a} + v\hat{a}^\dagger) \sum_{n=0}^{\infty} c_n |n\rangle \\ &= u \sum_{n=1}^{\infty} c_n \sqrt{n} |n-1\rangle + v \sum_{n=0}^{\infty} c_n \sqrt{n+1} |n+1\rangle \\ &= uc_1 |0\rangle + \sum_{n=1}^{\infty} (uc_{n+1} \sqrt{n+1} + vc_{n-1} \sqrt{n}) |n\rangle. \end{aligned} \quad (2.4.24)$$

Each term must vanish, so

$$\begin{cases} uc_1 = 0 \\ uc_{n+1} \sqrt{n+1} + vc_{n-1} \sqrt{n} = 0 \quad \text{for } n \geq 1. \end{cases} \quad (2.4.25)$$

The first condition implies $c_1 = 0$, while the second condition gives the iterative formula for $n \geq 2$:

$$c_n = -\frac{v}{u} \sqrt{\frac{n-1}{n}} c_{n-2}. \quad (2.4.26)$$

This means that all the odd terms must vanish:

$$c_{2k+1} = 0 \quad (2.4.27)$$

for $k \in \mathbb{N}$, while the even coefficients are

$$c_{2k} = \left(-\frac{v}{u}\right)^k \sqrt{\frac{(2k-1)!!}{(2k)!!}} c_0. \quad (2.4.28)$$

Therefore,

$$\begin{aligned} |\tilde{0}\rangle &= \sum_{k=0}^{\infty} c_{2k} |2k\rangle \\ &= \sum_{k=0}^{\infty} \left(-\frac{v}{u}\right)^k \sqrt{\frac{(2k-1)!!}{(2k)!!}} \frac{(\hat{a}^\dagger)^{2k}}{(2k)!} |0\rangle \\ &= \sum_{k=0}^{\infty} \left(-\frac{v}{u}\right)^k \frac{1}{(2k)!!} (\hat{a}^\dagger)^{2k} |0\rangle \\ &= \left\{ \sum_{k=0}^{\infty} \frac{1}{k!} \left[-\frac{v}{2u} (\hat{a}^\dagger)^2\right]^k \right\} |0\rangle \\ &= \exp\left(-\frac{v}{2u} \hat{a}^\dagger \hat{a}^\dagger\right) |0\rangle. \end{aligned} \quad (2.4.29)$$

Note that the new vacuum is not the old vacuum. It contains indefinite, even number of the original bosons.

Exercise 2.5

(a) Let $u_\ell = \phi_{2k}$, $v_\ell = \phi_{2k+1}$. The equation of motion is

$$\begin{cases} m\ddot{u}_\ell = -k(2u_\ell - v_{\ell-1} - v_\ell) \\ M\ddot{v}_\ell = -k(2v_\ell - u_\ell - u_{\ell+1}). \end{cases} \quad (2.5.1)$$

Transforming into the Fourier basis

$$\begin{cases} U_q = \frac{1}{\sqrt{N}} \sum_{\ell} e^{-2i\ell a q} u_{\ell} \\ V_q = \frac{1}{\sqrt{N}} \sum_{\ell} e^{-2i\ell a q} v_{\ell}, \end{cases} \quad (2.5.2)$$

where q lives in the first Brillouin zone $q \in (-\frac{\pi}{2a}, \frac{\pi}{2a}]$, the equation of motion is

$$\begin{cases} m\ddot{U}_q = -2kU_q + 2k \cos(aq)V_q \\ M\ddot{V}_q = -2kV_q + 2k \cos(aq)U_q. \end{cases} \quad (2.5.3)$$

In matrix notation,

$$\begin{pmatrix} m & 0 \\ 0 & M \end{pmatrix} \begin{pmatrix} \ddot{U}_q \\ \ddot{V}_q \end{pmatrix} = - \begin{pmatrix} 2k & -2k \cos(qa) \\ -2k \cos(qa) & 2k \end{pmatrix} \begin{pmatrix} U_q \\ V_q \end{pmatrix}, \quad (2.5.4)$$

or

$$\mathbf{M}\ddot{\mathbf{U}}_q = -\mathbf{K}_q \mathbf{U}_q. \quad (2.5.5)$$

Using the ansatz $\mathbf{U}_q(t) = \mathbf{A}_q e^{i\omega_q t}$, this is

$$\omega_q^2 \mathbf{M} \mathbf{A}_q = \mathbf{K}_q \mathbf{A}_q. \quad (2.5.6)$$

This is a generalised eigenvalue equation, which is not the easiest to solve. This inspires us to introduce the mass-scaled coordinates

$$\mathbf{X}_q = \begin{pmatrix} X_q \\ Y_q \end{pmatrix} = \begin{pmatrix} \sqrt{m} U_q \\ \sqrt{M} V_q \end{pmatrix} \quad (2.5.7)$$

such that

$$\mathbf{U}_q = \mathbf{M}^{-1/2} \mathbf{X}_q. \quad (2.5.8)$$

Then we have

$$\ddot{\mathbf{X}} = -\mathbf{D}_q \mathbf{X}_q \quad (2.5.9)$$

where

$$\mathbf{D} = \mathbf{M}^{-1/2} \mathbf{K} \mathbf{M}^{-1/2} = \begin{pmatrix} \frac{2k}{m} & -\frac{2k \cos(qa)}{\sqrt{mM}} \\ -\frac{2k \cos(qa)}{\sqrt{mM}} & \frac{2k}{M} \end{pmatrix}. \quad (2.5.10)$$

Now suppose $\mathbf{X}_q(t) = \mathbf{A}_q e^{i\omega_q t}$, we have the eigenvalue equation

$$\mathbf{D}_q \mathbf{A}_q = \omega_q^2 \mathbf{A}_q. \quad (2.5.11)$$

Solving for the eigenvalue equation

$$\det(\mathbf{D}_q - \omega_q^2 \mathbf{I}) = 0 \quad (2.5.12)$$

gives the two eigenvalues

$$\omega_{\pm}^2(q) = k \left(\frac{1}{m} + \frac{1}{M} \right) \pm k \sqrt{\left(\frac{1}{m} + \frac{1}{M} \right)^2 - \frac{4}{mM} \sin^2(qa)}. \quad (2.5.13)$$

We denote the two normalised eigenvectors

$$\mathbf{e}_{\pm}(q) = \begin{pmatrix} e_{\pm,1} \\ e_{\pm,2} \end{pmatrix}. \quad (2.5.14)$$

They satisfy

$$\left(\frac{2k}{m} - \omega_{\pm}^2 \right) e_{\pm,1} - \frac{2k \cos(qa)}{\sqrt{mM}} e_{\pm,2} = 0, \quad (2.5.15)$$

which gives

$$\frac{e_{\pm,2}}{e_{\pm,1}} = \frac{\sqrt{mM} \left(\frac{2k}{m} - \omega_{\pm}^2 \right)}{2k \cos(qa)}. \quad (2.5.16)$$

This should be solved along with the normalisation condition

$$e_{\pm,1}^2 + e_{\pm,2}^2 = 1, \quad (2.5.17)$$

but I do not bother to.

Let

$$\mathbf{E}_q = \begin{pmatrix} | & | \\ \mathbf{e}_+(q) & \mathbf{e}_-(q) \\ | & | \end{pmatrix}, \quad (2.5.18)$$

then

$$\mathbf{E}_q^T \mathbf{D}_q \mathbf{E}_q = \begin{pmatrix} \omega_+^2(q) & 0 \\ 0 & \omega_-^2(q) \end{pmatrix} =: \Omega_q, \quad (2.5.19)$$

and so

$$\ddot{\mathbf{X}}_q = -\mathbf{E}_q \Omega_q \mathbf{E}_q^T \mathbf{X}_q. \quad (2.5.20)$$

Therefore

$$\frac{d^2}{dt^2} (\mathbf{E}_q^T \mathbf{X}_q) = -\Omega_q (\mathbf{E}_q^T \mathbf{X}_q). \quad (2.5.21)$$

This is a diagonal equation of motion, and

$$\mathbf{Q}_q = \mathbf{E}_q^T \mathbf{X}_q \quad (2.5.22)$$

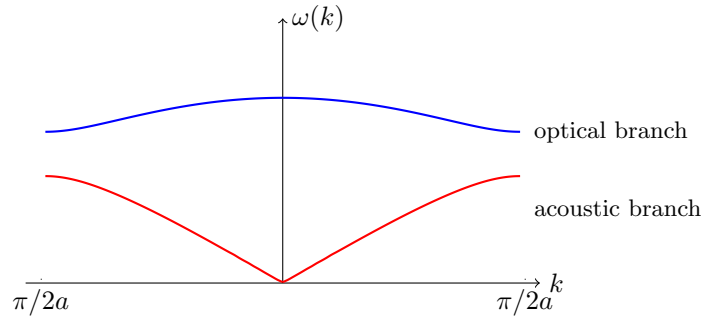
are the normal coordinates. The two normal coordinates are

$$Q_{q,\pm} = \mathbf{e}_{q,\pm}^T \mathbf{X}_q. \quad (2.5.23)$$

They satisfy

$$\ddot{Q}_{q,\pm} = -\omega_{q,\pm}^2 Q_{q,\pm}, \quad (2.5.24)$$

so that each is oscillating independently with frequency $\omega_{q,\pm}$.



(b) At the band edge $q = \frac{\pi}{2a}$,

$$\omega_{\pm}^2 = \frac{k}{mM} [(m+M) \pm (m-M)], \quad (2.5.25)$$

so

$$\begin{cases} \omega_+ = \sqrt{\frac{2k}{M}} \\ \omega_- = \sqrt{\frac{2k}{m}} \end{cases}. \quad (2.5.26)$$

The band gap is

$$\Delta\omega = \sqrt{2k} \left| \frac{1}{\sqrt{M}} - \frac{1}{\sqrt{m}} \right|. \quad (2.5.27)$$

(c) We will denote the momentum conjugate to $\mathbf{X}_q$ as $\mathbf{P}$, and that to $\mathbf{Q}_q$ as $\mathbf{\Pi}_q$. The kinetic energy is

$$\begin{aligned}
 T &= \frac{1}{2} \sum_q m \left| \dot{\mathbf{U}}_q \right|^2 + M \left| \dot{\mathbf{V}}_q \right|^2 \\
 &= \frac{1}{2} \sum_q \dot{\mathbf{X}}_q^\dagger \dot{\mathbf{X}}_q \\
 &= \frac{1}{2} \sum_q \mathbf{P}_q^\dagger \mathbf{P}_q \\
 &= \frac{1}{2} \sum_q \mathbf{\Pi}_q^\dagger \mathbf{\Pi}_q,
 \end{aligned} \tag{2.5.28}$$

and the potential energy is

$$\begin{aligned}
 V &= \frac{1}{2} \sum_q \mathbf{X}_q^\dagger \mathbf{D}_q \mathbf{X}_q \\
 &= \frac{1}{2} \sum_q \mathbf{Q}_q^\dagger \mathbf{\Omega}_q \mathbf{Q}_q.
 \end{aligned} \tag{2.5.29}$$

The Hamiltonian is therefore

$$H = \sum_q \sum_{\sigma=\pm} \left[\frac{1}{2} |\Pi_{q,\sigma}|^2 + \frac{1}{2} \omega_{q,\sigma}^2 |Q_{q,\sigma}|^2 \right]. \tag{2.5.30}$$

Quantising:

$$\hat{a}_{q,\sigma} = \sqrt{\frac{\omega_{q,\sigma}}{2\hbar}} Q_{q,\sigma} + \frac{i}{\sqrt{2\hbar\omega_{q,\sigma}}} \Pi_{q,\sigma} \tag{2.5.31}$$

such that

$$[\hat{a}_{q,\sigma}, \hat{a}_{q',\sigma'}^\dagger] = \delta_{qq'} \delta_{\sigma\sigma'}. \tag{2.5.32}$$

The Hamiltonian is

$$\hat{H} = \sum_q \sum_{\sigma=\pm} \left[\hbar\omega_{q,\sigma} \left(\hat{a}_{q,\sigma}^\dagger \hat{a}_{q,\sigma} + \frac{1}{2} \right) \right]. \tag{2.5.33}$$

Exercise 2.6

(a) The RMS displacement

$$\Delta r_{\text{RMS}} = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}, \tag{2.6.1}$$

where in each direction,

$$\Delta x \sim \sqrt{\frac{\hbar}{m\omega}} \tag{2.6.2}$$

by the uncertainty principle. Therefore, by Linderman criterion, the crystal is unstable if

$$\Delta r_{\text{RMS}} \sim \sqrt{\frac{\hbar}{m\omega}} \gtrsim \frac{1}{3}a \implies \frac{\hbar}{m\omega a^2} > \zeta_c \tag{2.6.3}$$

for some dimensionless ζ_c .

(b) Have

$$\begin{aligned}\hat{\Phi}_j &= \frac{1}{\sqrt{N_S}} \sum_{\mathbf{q}} e^{i\mathbf{q} \cdot \mathbf{R}_j} \hat{\Phi}_{\mathbf{q}} \\ &= \sqrt{\frac{\hbar}{2mN_S}} \sum_{\mathbf{q}} e^{i\mathbf{q} \cdot \mathbf{R}_j} \frac{1}{\sqrt{\omega_{\mathbf{q}}}} (\hat{a}_{\mathbf{q}} + \hat{a}_{-\mathbf{q}}^\dagger).\end{aligned}\quad (2.6.4)$$

Take $j = 0$,

$$\hat{\Phi} = \sqrt{\frac{\hbar}{2mN_S}} \sum_{\mathbf{q}} \frac{1}{\sqrt{\omega_{\mathbf{q}}}} (\hat{a}_{\mathbf{q}} + \hat{a}_{-\mathbf{q}}^\dagger). \quad (2.6.5)$$

To work out $\omega_{\mathbf{q}}$, consider

$$\Phi_{\mathbf{R}+\mathbf{a}} - \Phi_{\mathbf{R}} = \frac{1}{\sqrt{N_S}} \sum_{\mathbf{q}} \Phi_{\mathbf{q}} e^{i\mathbf{q} \cdot \mathbf{R}} (e^{i\mathbf{q} \cdot \mathbf{a}} + 1). \quad (2.6.6)$$

By Fourier orthogonality

$$\sum_{\mathbf{R}} e^{i(\mathbf{q}+\mathbf{q}') \cdot \mathbf{R}} = N \delta_{\mathbf{q}', -\mathbf{q}}, \quad (2.6.7)$$

we have

$$\begin{aligned}\sum_{\mathbf{R}} (\Phi_{\mathbf{R}+\mathbf{a}} - \Phi_{\mathbf{R}})^2 &= \sum_{\mathbf{q}} \Phi_{\mathbf{q}} \Phi_{-\mathbf{q}} (e^{i\mathbf{q} \cdot \mathbf{a}} - 1)(e^{-i\mathbf{q} \cdot \mathbf{a}} - 1) \\ &= \sum_{\mathbf{q}} 4 \sin^2 \left(\frac{\mathbf{q} \cdot \mathbf{a}}{2} \right) \Phi_{\mathbf{q}}^\dagger \Phi_{\mathbf{q}}.\end{aligned}\quad (2.6.8)$$

Summing over the three directions $\mathbf{a} \in \{a\hat{\mathbf{x}}, a\hat{\mathbf{y}}, a\hat{\mathbf{z}}\}$,

$$V = \frac{1}{2} m \omega^2 \sum_{\mathbf{q}} 4 \sin \left[\sin^2 \frac{q_x a}{2} + \sin^2 \frac{q_y a}{2} + \sin^2 \frac{q_z a}{2} \right] \Phi_{\mathbf{q}}^\dagger \Phi_{\mathbf{q}}, \quad (2.6.9)$$

and hence

$$\omega_{\mathbf{q}} = 2\omega \sqrt{\sin^2 \frac{q_x a}{2} + \sin^2 \frac{q_y a}{2} + \sin^2 \frac{q_z a}{2}}. \quad (2.6.10)$$

Therefore,

$$\begin{aligned}\langle 0 | \hat{\Phi}^2 | 0 \rangle &= \frac{\hbar}{2mN_S} \sum_{\mathbf{q}, \mathbf{q}'} \frac{1}{\sqrt{\omega_{\mathbf{q}} \omega_{\mathbf{q}'}}} \langle 0 | (\hat{a}_{\mathbf{q}} + \hat{a}_{-\mathbf{q}}^\dagger) (\hat{a}_{\mathbf{q}'} + \hat{a}_{-\mathbf{q}'}^\dagger) | 0 \rangle \\ &= \frac{\hbar}{2mN_S} \sum_{\mathbf{q}, \mathbf{q}'} \frac{1}{\sqrt{\omega_{\mathbf{q}} \omega_{\mathbf{q}'}}} \underbrace{\langle 0 | \hat{a}_{\mathbf{q}} \hat{a}_{-\mathbf{q}'}^\dagger | 0 \rangle}_{\delta_{\mathbf{q}, -\mathbf{q}'}} \\ &= \frac{\hbar}{2mN_S} \sum_{\mathbf{q}} \frac{1}{\omega_{\mathbf{q}}}.\end{aligned}\quad (2.6.11)$$

We will approximate the sum by an integral

$$\frac{1}{N_S} \sum_{\mathbf{q}} \rightarrow \left(\frac{a}{2\pi} \right)^3 \int_{-\pi/a}^{\pi/a} d^3 \mathbf{q}. \quad (2.6.12)$$

Then

$$\begin{aligned}\langle 0 | \hat{\Phi}^2 | 0 \rangle &= \frac{\hbar}{4\pi\omega} \int_{-\pi}^{\pi} \frac{-\pi}{\pi} \frac{d^3 \mathbf{x}}{(2\pi)^3} \frac{1}{\sqrt{\sin^2(\frac{x}{2}) + \sin^2(\frac{y}{2}) + \sin^2(\frac{z}{2})}} \\ &= \frac{\hbar}{4m\omega} I,\end{aligned}\quad (2.6.13)$$

where

$$I = 0.910688 \dots \quad (2.6.14)$$

For the crystal to be unstable,

$$\langle 0 | \hat{\Phi}^2 | 0 \rangle > \frac{\hbar}{4m\omega} I > \frac{1}{9} a^2, \quad (2.6.15)$$

and so

$$\frac{\hbar}{m\omega a^2} > \frac{4}{9I} =: \zeta_c \approx 0.4880 \dots \quad (2.6.16)$$

(c) Under the unrealistic assumption that ω is independent of a , Helium is *stable* if

$$\frac{\hbar}{m\omega a^2} < \zeta_c, \quad (2.6.17)$$

giving

$$a > a_c = \sqrt{\frac{\hbar}{m\omega\zeta_c}} \approx 2.878 \times 10^{-11} \text{ m}. \quad (2.6.18)$$

This is apparently unrealistic since crystalline Helium should be *unstable* at low density (for a greater than some a_c).

Using the realistic assumption that $\omega \sim a^{-\alpha}$ for some $\alpha > 2$, we now have that Helium is *stable* if

$$\frac{\hbar}{m\omega a^2} = \frac{\hbar}{m\omega_0} a^\gamma < \zeta_c \quad (2.6.19)$$

for some $\gamma = \alpha - 2 > 0$. This is

$$a < \left(\frac{\zeta_c m \omega_0}{\hbar} \right)^{1/\gamma} =: a_c. \quad (2.6.20)$$